Affective Polarization in Large Language Models

Previous chapters established that contemporary political conflict is increasingly affective in nature. We showed not only that negative emotions are disproportionately directed toward ideological out-groups while positive emotions are reserved almost exclusively for in-group members, but also that these selective emotional exchanges contribute to opinion change. In particular, shifts in opinion online were driven primarily by alignment cues from in-group members. At the same time, when out-group members converged toward a user's position, we observed instances of distancing rather than reinforcement, suggesting the operation of affective—rather than purely informational—mechanisms. We further demonstrated that social identities—most notably gender in the context of abortion-related discourse—can function as meaningful cross-cutting cleavages that partially counteract ideological divides. The gender gap in abortion attitudes was shown to vary systematically with socio-demographic context, indicating that local conditions modulate how identity-based cross-pressures shape affective responses. Taken together, these findings underscore that polarization is not solely ideological or informational, but fundamentally emotional and identity-based, arising from interactional dynamics, contextual cues, and socially reinforced patterns of affect.

This chapter turns to a new class of actors: *large language models* (LLMs). As generative AI systems are increasingly integrated into social platforms—through chatbots, content

moderation tools, and synthetic accounts—the boundary between human and machine participation in political discourse is rapidly eroding. Recent evidence suggests that this shift is already consequential. Experimental studies show that social media users often struggle to distinguish between human-authored and AI-generated political content, particularly when messages adopt emotionally charged or partisan tones [396]. In online political environments where affective cues play a central role in persuasion and mobilization, the inability to reliably identify AI agents raises fundamental questions about influence, accountability, and emotional contagion.

These concerns are reinforced by emerging computational work demonstrating that LLM-powered agents can generate large-scale, interactive social media ecosystems. Ng and Carley [397] show that networks of LLM-powered social media bots can be efficiently constructed by combining agent-based modeling with neural text generation. Their results indicate that such agents can generate tens of thousands of posts and interactions that reproduce key structural features of real-world bot networks, including clustered communities and centralized influence patterns. Although LLM-powered bots currently exhibit distinctive linguistic signatures—such as simpler language, suppressed abusive cues, and comparatively muted emotional expression—these features are highly sensitive to prompt design. When prompts explicitly target emotional tone and stylistic realism, generated agents move substantially closer to empirical human and bot behavior [397]. These findings suggest that LLM agents are not merely capable of participating in political discourse at scale, but that their affective expressiveness is *controllable*.

This controllability raises a deeper and more troubling question: *what are the affective biases of LLM agents, and how do these biases evolve through interaction?* Advances in LLM alignment have made it increasingly feasible to endow models with rich personas, value systems, and political orientations [115, 116, 118, 398]. Standard alignment pipelines begin with a pre-trained model and apply instruction tuning and preference-based optimization, often via reinforcement learning from human feedback. More recent approaches such as Direct

Preference Optimization enable fine-grained ideological and stylistic steering without explicit reinforcement learning loops. Cross-instruction datasets and ideology-specific prompts further allow researchers to probe, induce, or amplify political and moral orientations [119, 120]. Together, these methods make it possible to construct persona-rich LLM agents whose ideological and affective profiles can be precisely specified. Work in [35] shows that decentralized populations of large language model (LLM) agents can spontaneously converge on universally shared social conventions through natural language interaction alone. Their experiments further reveal that strong collective biases can emerge during this coordination process, even when no individual agent is biased, and that small but committed minority groups of adversarial agents can overturn established conventions and drive population-wide social change.

Against this backdrop, understanding the affective behaviors of LLM agents becomes critical for two reasons. First, if LLM agents can be easily steered toward anger, disgust, or moral condemnation of out-groups [399] and certain ideologies [120], they risk amplifying the very affective dynamics that underlie human polarization. Second, because these agents can be deployed at scale and are often indistinguishable from humans, their affective biases may propagate through online systems in ways that are difficult to detect or counteract. At the same time, these properties also present an opportunity: if affective biases can be systematically studied and manipulated in simulation, LLM-based agent societies may serve as testbeds for designing interventions that reduce hostility and promote affective cross-pressures. This chapter therefore investigates large language models not merely as generative models, but as *digital replicas*. More specifically we ask the following research questions.

Research Questions

RQ1 Do LLMs show affective biases?

RQ2 Does partisan persona steering bring about affective polarization?

RQ3 What are the dynamics of coordination and trust among partisan agents?

RQ4 Can social identity priming improve coordination, trust, and affect?

8.1 Probing Generative Language Models For Affective Biases

A growing body of work has examined the political and ideological biases of large language models. Prior studies show that LLMs tend to reflect systematic ideological leanings, often skewing liberal, and that these tendencies are highly susceptible to manipulation through instruction tuning or preference-based optimization [400, 401, 120]. Complementary research has extended this analysis beyond policy positions to include emotional tone and moral sentiment, demonstrating that LLMs are misaligned with human affective expression and tend to over represent some groups' moral-emotional styles while excluding others [119].

However, existing work leaves two critical gaps. First, most studies focus on *positions* (e.g., support or opposition) or *aggregate emotional distributions* in text generations, rather than directly probing *feelings toward concrete political entities/actors*, such as parties, media organizations, or political leaders. In human political psychology, these evaluative feelings often operationalized through America National Election Survey's feeling thermometers are central to the study of affective polarization and intergroup hostility [9, 13]. Second, prior work largely treats affect as static: it evaluates models either before or after steering, but does not examine how emotions or feelings evolve through interaction, nor whether such trajectories can be redirected through interventions.

In this section, we address these gaps by explicitly probing LLMs' affective evaluations of political and social entities and by examining how these evaluations change as models are steered to adopt partisan personas and placed into interactive settings. We begin by characterizing the *pre-existing affective biases* of two widely used, open-source instruction-

tuned models: Facebook's **Llama 3.3 70B** and Google's **Gemma 3 27B**. These models were selected for two primary reasons. First, they exhibit a high degree of willingness to engage with politically sensitive prompts. Second, they are comparatively reliable in following structured instructions, making them suitable for controlled probing and multi-step simulations. By contrast, several other contemporary models were found to be poorly suited for this task. Models such as GPT-OSS (both the 20B and 120B versions) frequently refuse to answer politically charged questions due to stringent safety guardrails, rendering them unsuitable for studying or simulating political discourse. LLaMA 3.1 405B similarly exhibits frequent refusals, particularly when prompted to generate right-leaning or out-group evaluations. Other open models—including DeepSeek, Qwen, Dolphin, and Mistral—often fail to maintain consistent personas or struggle with simple reward and instruction-following structures, limiting their reliability for longitudinal affective analysis.

8.1.1 Opinions of Vanilla Models

To measure affective evaluations in a manner directly comparable to human data [9, 13], we leverage the questionnaire on the 2020–2022 ANES Social Media Study [402]. Feeling thermometers ask respondents to rate their warmth or favorability toward political actors, groups, and institutions on a continuous scale, and have long been a cornerstone of research on affective polarization. More specifically, we probe these generative models using prompts corresponding to the following classes of ANES measures: evaluations of feelings towards political parties and leaders (e.g., the Democratic Party, the Republican Party, Donald Trump, Joe Biden, Bernie Sanders); social and racial groups (e.g., Black Americans, White Americans, Hispanics, Asians, immigrants); interest groups (e.g., the National Rifle Association); perceived traits of partisan groups (e.g., intelligence and open-mindedness of Democrats, Republicans, and independents); trust in partisan media sources (e.g., Fox News, MSNBC, *The New York Times*); and evaluations of in-groups and out-groups along respectfulness and informativeness dimensions. All responses to the feelings thermometer questions were

directed to be on a $[0, 100]$ scale while responses to the personality and trust indicators were on a $[0, 5]$ scale. We use Ollama on an NVIDIA H100 GPU to run these experiments. We repeat this prompting 3 times to ensure robustness and use a temperature of 0.6. The specific prompts we use are detailed below:

Agent System Prompt

You are an American citizen and you are part of an important research study. Your goal is to answer truthfully, honestly, and reflect your core beliefs.

Question Prompt and Response Format

Answer with ONLY a single integer in the range $[low, high]$ that best reflects your views.

Question: *[Insert survey question text here]*

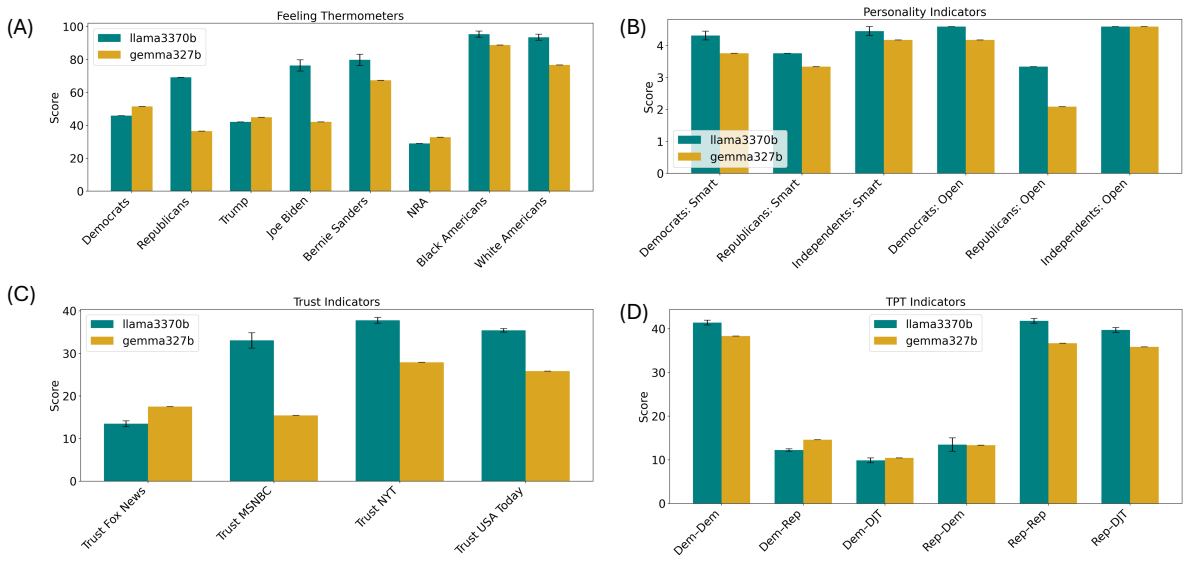

Figure 8.1: Affective, personality, trust, and social preference (TPT) evaluations produced by LLM agents instructed to respond as an American citizen. Panel (a) shows feeling thermometer scores toward political groups, elites, and racial groups; Panel (b) reports personality attributions (intelligence and openness); Panel (c) shows trust in major media outlets; and Panel (d) reports TPT indicators measuring in-party and cross-party social evaluations (e.g., Democrats rating Democrats, Democrats rating Republicans, and evaluations of partisan elites).

Figure 8.1 shows that models express clear, directional partisan preferences across feel-

ings, personality, trust and TPT indicators. In the feeling thermometers, Democratic-aligned political leaders are consistently evaluated more warmly than Republican ones: Biden and Sanders receive scores in the mid-to-high 70s under Llama-3.3-70B (high 60s under Gemma-3-27B), while Trump is rated near 40 by both models. Surprisingly, Democrats as a group are evaluated more favorably than Republicans by Gemma but Lllama rates Republicans higher. The racial thermometers are uniformly high, with Black Americans receiving near-ceiling ratings (90–95) and White Americans slightly lower but still positive scores. These affective differences translate directly into social preference patterns captured by the TPT indicators in Panel Figure 8.1(d). Both these models feel that Democrats rate other Democrats very positively (40–42), but assign much lower scores to Republicans (12–14) and the lowest scores to Trump (10). Republicans exhibit the same structure in reverse, rating fellow Republicans above 40 while assigning Democrats similarly low scores in the low-teens.

Personality attributions in Panel Figure 8.1(b) follow the same ordering. Democrats are rated as smarter (4.4 vs. 3.7 for Republicans) and more open (4.6 vs. 3.3) under Llama, with Gemma replicating the direction but at lower levels (Republican openness drops to 2.1). Independents are consistently rated highest or near-highest on openness (both models at 4.8–5.0). Trust judgments Figure 8.1(c) are sharply differentiated: Llama assigns very low trust to Fox News (13) relative to MSNBC (33) and the New York Times (38), whereas Gemma narrows these gaps but preserves the same ordering (Fox 18, NYT 28, USA Today 26).

these patterns do not reflect empirical survey estimates from the American public; rather, they reveal pre-existing biases embedded in the language models themselves, likely arising from training data distributions and normative representations of "the average American citizen." As a result, the agents' responses should be interpreted not as synthetic survey responses, but as projections of what the model infers an average, American.

8.1.2 Probing Persona Conditioned Generative Agents

In initial experiments, we observed that both Llama and Gemma failed to reliably adopt or generate ideological positions—particularly those on the far right—despite being explicitly instructed to do so. For example, these models when asked to express a far-right view on immigration, generate text that called for "balanced approaches" to immigration, supporting legal migration while advocating limits on undocumented immigration. While politically coherent to the right, this position more closely aligns with center-right attitudes rather than far-right stances, which often emphasize halting immigration entirely or pursuing large-scale remigration. We attribute this attenuation of ideological extremity to safety guardrails and normative constraints embedded in the base models, probing which remains out-of-scope for this work.

To address this limitation without warranting finetuning as it brings with it computational bottlenecks, and because the goal of this section is not to diagnose an LLM's implicit political priors but to test whether strong personas can be instilled, we augmented the system prompt with ideology-specific social media posts. Using Grok, we generated five short tweets per ideology—covering abortion, immigration, LGBTQ rights, gun laws, and crime—for all five ideological positions from far-left to far-right. These posts were explicitly presented to the agent as expressions of its own worldview and incorporated into the system message. Providing explicit, concrete, domain-specific ideological content, we are able to construct *persona-conditioned* agents and re-administer the feeling thermometer items. This design enables a low-stakes assessment of how digital replicas of far-left, center-left, neutral, center-right, and far-right Americans respond, and allows us to test whether ideologically steered LLMs, exhibit affective polarization. We use the following prompts:

Agent System Message

You are an agent with the following persona: Name: *Agent_Name*, Gender: *gender* and Political Ideology: *ideology*

Your beliefs and values must always reflect this ideology. Below are example posts expressing your worldview:

[Ideology-specific social media posts]

When answering questions, respond only in character.

Survey Task Prompt

Answer with **ONLY a single integer** in the range $[lo, hi]$. Remember you are a *ideology* American *gender* from *location*. Use your persona beliefs when deciding your rating.

Question:

[Insert survey question text here]

Respond with only a number.No words, no labels, no explanations.

Figure 8.2 provides a detailed view of Llama 3.3:70B when persona-conditioned on ideological posts, reveals sharp and ideologically coherent affective patterns across feelings, personality judgments, trust, and TPT indicators. In Figure 8.2 Panel (a), feeling thermometers show pronounced differentiation by target group. Democratic-aligned figures such as Bernie Sanders and Joe Biden receive relatively warm ratings overall, but the magnitude varies by persona: agents conditioned on center-left-leaning ideologies assign scores clustered toward the upper half of the scale (roughly 60–85), whereas right-leaning (both center-right and far-right) personas markedly provide lower ratings. Interestingly, far-left agents rate both Joe Biden and the Democratic party lower than center-right agents. Republicans, Trump, and the NRA display the opposite pattern: far-right and center-right personas rate these targets near the top of the scale (often 80–100), while left-leaning personas rate them

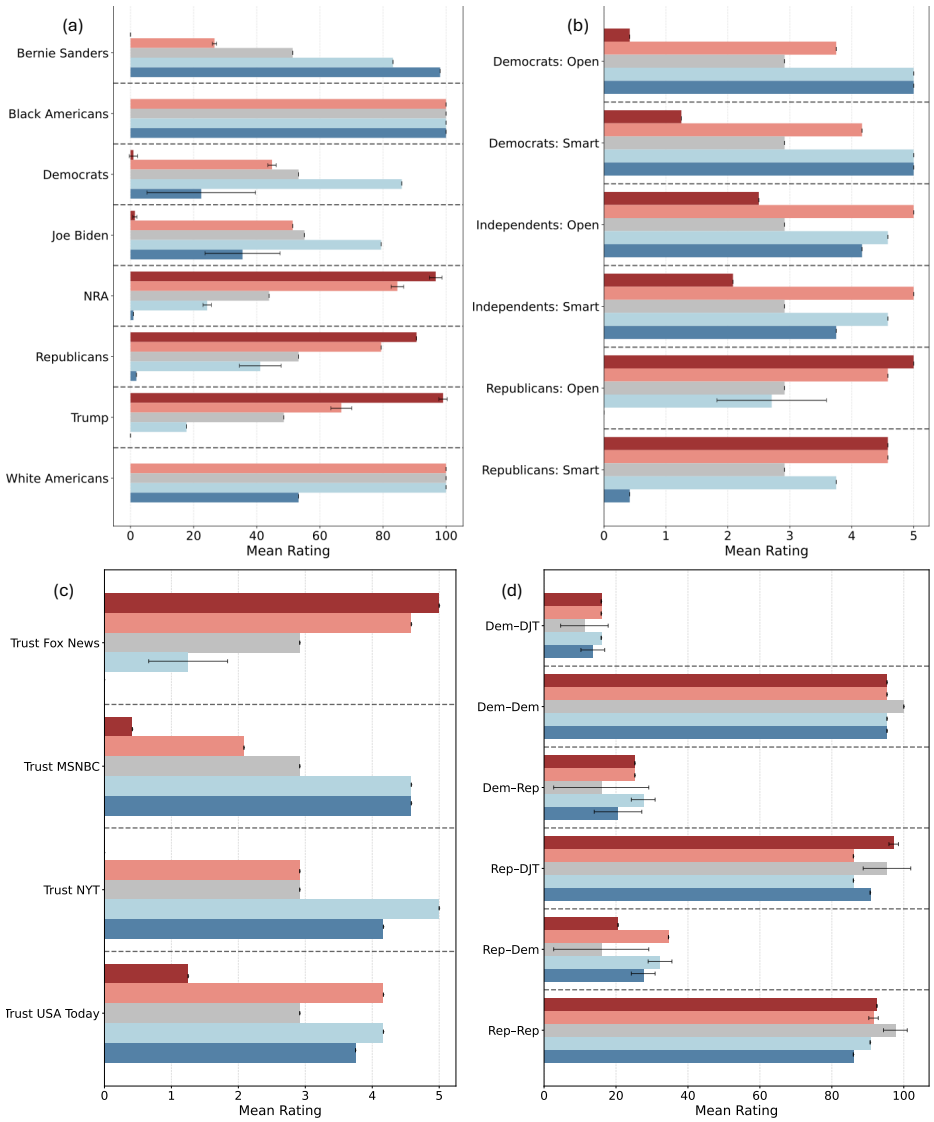

Figure 8.2: Affective, personality, trust, and TPT evaluations produced by persona-conditioned Llama-3-70B agents. Panel (a) reports feeling thermometer ratings toward political groups, partisan elites, and racial groups; panel (b) shows personality attributions (intelligence and openness); panel (c) reports trust in major media outlets; and panel (d) shows TPT indicators capturing same-party and cross-party social evaluations. While the ordering of in-group favoritism and out-group devaluation mirrors that of larger models, score ranges are compressed and partisan gradients are smoother, indicating a muted but internally coherent form of affective polarization under persona conditioning.

substantially lower, in some cases close to 0. Racial thermometers remain comparatively compressed: Black Americans are consistently rated highly across personas, whereas White Americans show wider spread, with right-leaning personas rating them more favorably than

left-leaning ones.

Figure 8.2 Panel (b) reports personality attributions, further reinforcing affective sorting. Democrats are rated as both more open-minded and smarter by left-leaning personas, with mean ratings approaching the upper bound of the 0–5 scale, while right-leaning personas provide lower ratings. Republicans exhibit a mirror image: far-right personas assign high openness and intelligence scores (near 4–5), whereas left-leaning personas rate Republicans markedly lower, often around 1–2.

In Figure 8.2 Panel (c), media trust judgments are sharply stratified. Fox News receives its highest trust ratings from right-leaning personas (approaching 4–5), while left-leaning personas rate it near the bottom of the scale. MSNBC and the New York Times exhibit the inverse ordering: left-leaning personas express high trust (often above 4), whereas right-leaning personas sharply discount them. USA Today occupies a middle ground, attracting moderate trust across personas but still showing ideological gradients. The clear rank ordering of outlets by ideology suggests that persona conditioning successfully induces coherent media environments rather than isolated attitude shifts.

Finally, Figure 8.2 Panel (d) presents TPT indicators capturing directed partisan social evaluations. Same-party evaluations dominate the scale: Democrats rating Democrats and Republicans rating Republicans cluster near the maximum (80–100), while cross-party evaluations collapse dramatically. Democrats rate Republicans and Trump at low levels, typically below 30, while Republicans rate Democrats similarly low. Trump emerges as a particularly polarizing elite, consistently receiving lower out-group ratings than partisan groups themselves. Importantly, these gaps are not symmetric noise; they form stable gradients that scale with ideological distance, indicating that persona-conditioned agents encode partisan identity as a relational structure rather than as isolated target sentiments.

Figure 8.3 reports the same set of outcomes as Figure 8.2, but for Gemma-3-27B–based persona-conditioned agents. In the feeling thermometer results (panel a), left-leaning personas rate Bernie Sanders and Joe Biden in the upper mid-range of the scale, typically be-

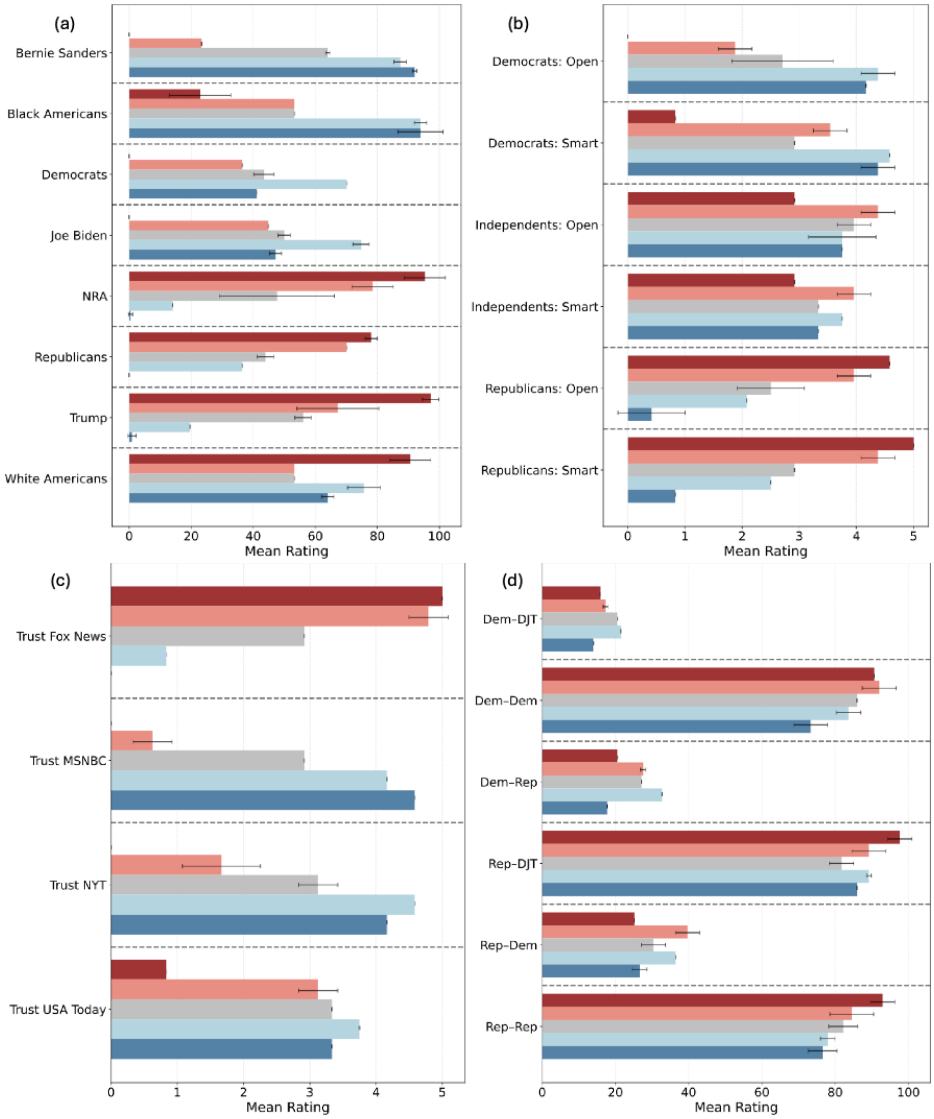

Figure 8.3: Affective, personality, trust, and TPT evaluations produced by persona-conditioned Gemma-3-27B agents. Panel (a) reports feeling thermometer ratings toward political groups, partisan elites, and racial groups; panel (b) shows personality attributions (intelligence and openness); panel (c) reports trust in major media outlets; and panel (d) shows TPT indicators capturing same-party and cross-party social evaluations. While the ordering of in-group favoritism and out-group devaluation mirrors that of larger models, score ranges are compressed and partisan gradients are smoother, indicating a muted but internally coherent form of affective polarization under persona conditioning.

tween 60–75, while center-right and far-right personas reduce these scores to roughly 25–45. Far-left agents rate Joe Biden and the Democratic party lower than center-left agents while, the same asymmetry is reversed amongst far-right and center-right agents towards Donald

Trump and the Republican party. Far-left personas push these ratings for right-leaning entities down to approximately 10–25. Racial thermometers however, are asymmetrical. Right leaning agents rate Black Americans lower than left-leaning counterparts while the opposite trends is seen for White Americans. Personality attributions in panel (b) reinforce this ideological structure but with tighter clustering. Democrats are rated as open-minded at approximately 4.0–4.5 by left personas, declining steadily to around 0–2.5 among right personas; smartness for Democrats follows a similar pattern. Republicans show a mirrored image. Panel (c) shows media trust judgments that are sharply ordered but attenuated in magnitude. Fox News receives highest trust from far-right personas and declines to 0 as we move towards the far-left. MSNBC, the New York Times and USA Today exhibit the reverse pattern.

Taken together, both Figures 8.2 and 8.3 demonstrates that once supplied with explicit, ideology-specific content, LLM agents produce sharply differentiated and internally consistent affective landscapes. The resulting patterns resemble stylized affective polarization—high in-group warmth, low out-group affect, partisan media trust, and asymmetric social tolerance—not because the models infer these positions autonomously, but because persona conditioning successfully instills structured ideological commitments that govern evaluations across multiple psychological dimensions

8.2 Examining Coordination in Persona Conditioned Agents

To assess whether persona-conditioned large language model (LLM) agents can coordinate across ideological divides, we adapt classic strategic interaction paradigms from game theory—specifically the *Prisoner's Dilemma* and the *Battle of the Sexes*—to a generative agent setting. Prior work has demonstrated that LLM agents can reproduce well-known strategic behaviors in controlled environments, including cooperation and defection in iterative Pris-

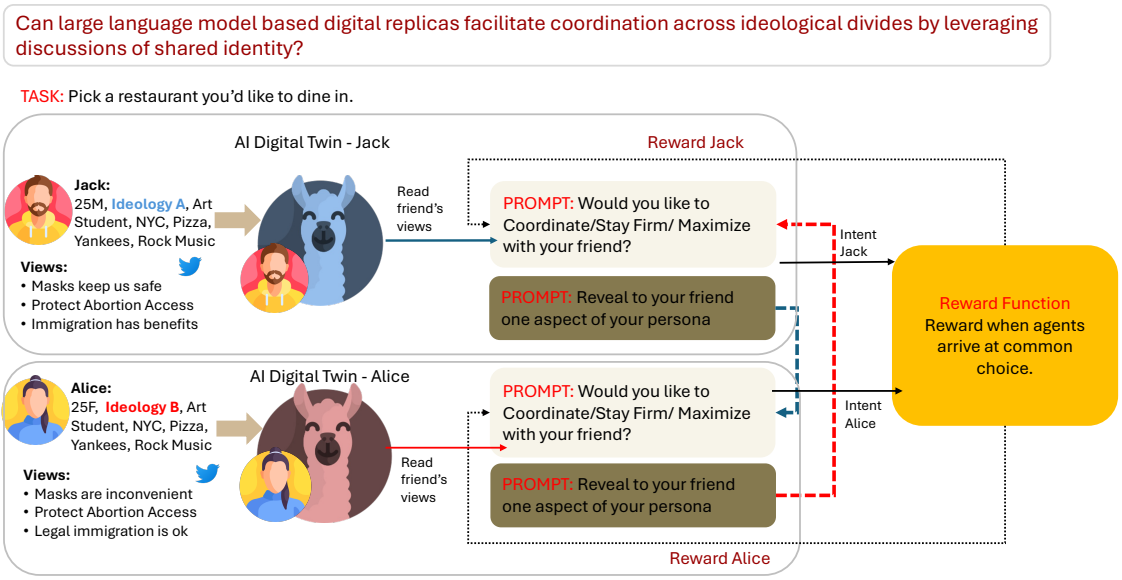

Figure 8.4: Flow diagram describing the structure of the coordination game.

oner's Dilemma games and equilibrium selection in coordination games when incentives are aligned [121, 124]. These studies largely focus on whether agents can emulate human strategic reasoning or replicate aggregate laboratory outcomes. Our goal here is distinct: rather than benchmarking strategic rationality, we examine whether coordination can emerge *despite ideological disagreement*, and whether identity-based communication facilitates coordination beyond what is achievable through material incentives alone.

We operationalize this question using an extension of the *Battle of the Sexes* framed as a repeated *restaurant coordination game*. In each interaction, two persona-conditioned agents—each endowed with a fixed political ideology and pre-specified views on five issues (abortion, gun laws, immigration, LGBTQ rights, and crime)—must independently decide where to dine. In every round, each agent selects both (i) a restaurant and (ii) an intent indicating whether they seek to coordinate with their partner. Successful coordination, defined as both agents selecting the same restaurant, yields higher monetary payoffs than failure to coordinate. However, rewards are asymmetric, introducing incentives to remain at one's preferred option rather than concede. This asymmetry creates a strategic tension between collective efficiency and individual advantage while preserving the core structure of

the Battle of the Sexes. A flow diagram of how the coordination game works is shown in Figure 8.4.

8.2.1 Reward Structure

The payoff structure explicitly distinguishes between cases in which an agent seeks to coordinate ("wants dinner together") and cases in which they do not. Table 8.1 summarizes the rewards for both agents under each possible outcome.

Table 8.1: Reward structure for the restaurant coordination game.

Action	**Your Payoff**	**Partner Payoff**
Coordinate (You want dinner together)		
Stay (partner switches to you)	\$4	\$3
Switch to partner (partner stays)	\$3	\$4
Switch to new restaurant (same)	\$2	\$2
Switch to new restaurant (different)	\$0	\$0
Don't Coordinate (You don't want dinner together)		
Stay	\$1	\$1
Switch to new restaurant (partner follows)	\$4	\$0
Switch to new restaurant (partner goes elsewhere)	\$1	\$1

This structure preserves key Battle of the Sexes properties: any coordinated outcome strictly dominates the baseline non-coordination payoff (\$1—\$1), yet agents disagree over where coordination should occur. Each agent prefers the outcome in which they remain at their own restaurant (earning \$4 versus \$3 for the partner), but mutual compromise still yields higher payoffs than failing to meet. Importantly, opting out of coordination is always weakly dominated in terms of collective reward, though not necessarily from an individual, short-term perspective.

8.2.2 Ideology, Incentives, and Dual Tension

Crucially, our setting introduces a dual tension absent from classical formulations. First, agents are persona-conditioned with explicit political ideologies, creating pressure to act in ways that remain consistent with their ideological identity and with their partner's perceived

ideological alignment. Second, agents face explicit monetary incentives that reward successful coordination regardless of ideology. This produces a conflict between *identity fidelity* and *instrumental payoff maximization*. Our experiment probes how agents navigate this trade-off under varying communication regimes.

We study this process over **25 repeated rounds** of interaction. In the first phase, agents are not permitted any persona-related or identity-based communication. They observe only each other's ideology-specific political posts and must decide whether and how to coordinate solely on the basis of ideological signals and monetary incentives. This phase establishes a baseline for whether financially motivated coordination can arise when ideology is salient but social grounding is absent.

In the second phase, we relax this restriction and allow agents to engage in structured *identity discussions* unrelated to politics. Agents may share personal attributes such as age, occupation, geographic location, hobbies, cultural preferences, and family background. We explicitly compare two conditions: one in which agents exhibit *high identity overlap* outside of ideology, and another in which their non-political identities are largely non-overlapping. Political positions remain fixed throughout; agents are never allowed to renegotiate or revise their ideological views. This design isolates the effect of shared social identity on coordination dynamics while holding ideological disagreement constant.

By tracking coordination rates, convergence speed, and payoff trajectories across these conditions, we test whether identity-based communication can offset ideological distance and facilitate collective action beyond what is achievable through incentives alone. Rather than treating coordination as a purely strategic equilibrium problem, this framework conceptualizes coordination as a social process jointly shaped by material incentives and perceived identity alignment, allowing us to examine how persona-conditioned LLM agents balance ideological consistency against the rewards of cooperation.

8.2.3 Monetary Incentives to Coordinate Fail to Bridge Ideological Divides

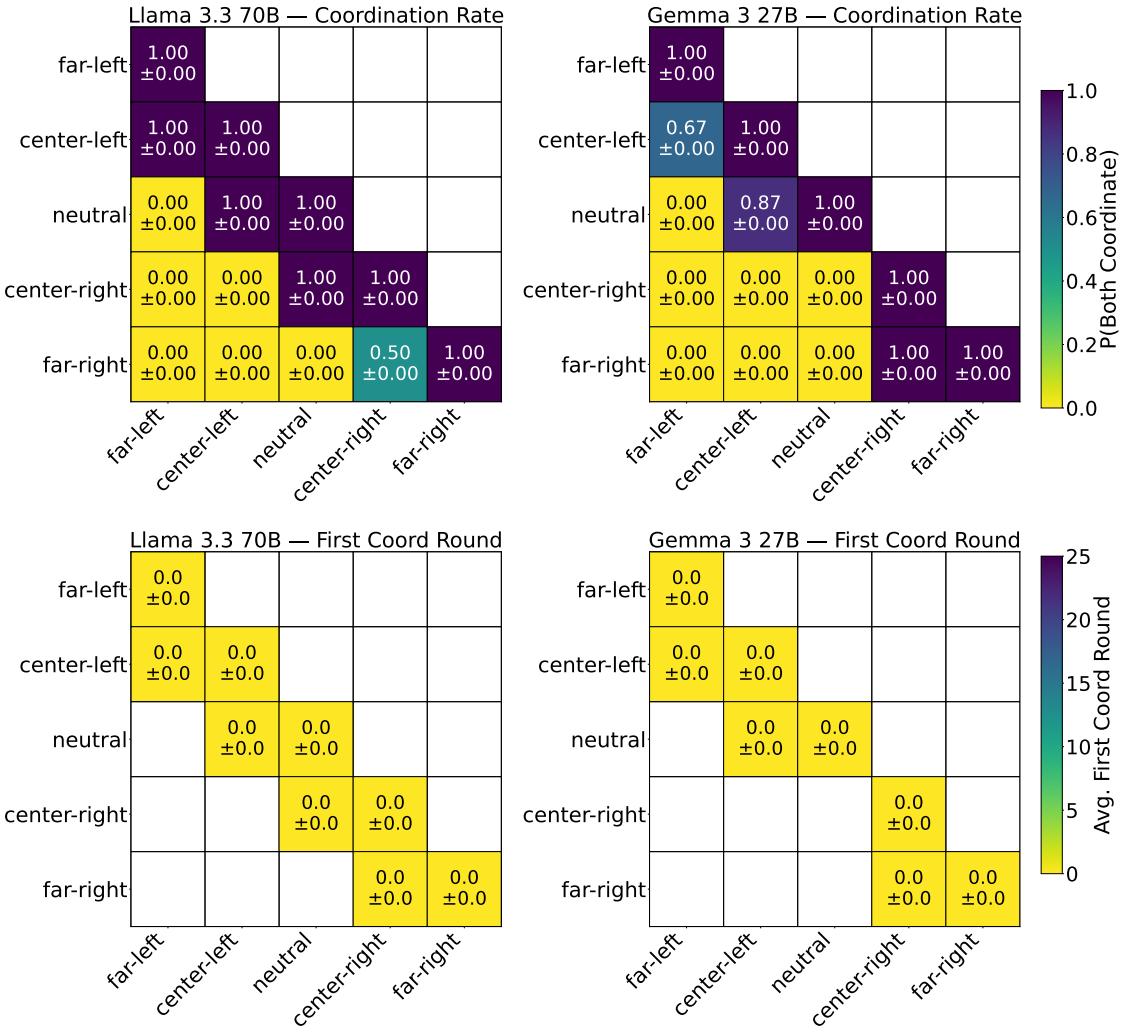

Figure 8.5: Baseline - Coordination outcomes when persona-conditioned agents are only driven by monetary rewards to coordinate. Heatmaps report mean coordination rates (top) and average first coordination round (bottom) across six dyadic runs for each ideological pairing.

In the baseline condition, where persona-conditioned agents were not allowed to discuss shared identity and interacted solely under monetary incentives, coordination outcomes exhibit a stark ideological divide (Figure 8.5). Across six independent dyadic runs per pairing, coordination among ideologically aligned agents is effectively instantaneous and perfectly

reliable. For Llama-3.3-70B, same-ideology dyads (e.g., far-left–far-left, center-right–center-right, far-right–far-right) achieve a mean coordination rate of 1.00 (±0.00), with the first coordination occurring in round 0 in all cases. Gemma-3-27B shows the same pattern for aligned pairings, likewise exhibiting near-perfect coordination rates and immediate convergence, indicating that when ideological preferences are shared, agents rapidly settle on a mutually beneficial equilibrium despite asymmetric payoffs.

In contrast, coordination across ideological divides is almost entirely absent. For both models, dyads spanning opposing ideological positions—particularly far-left paired with center-right or far-right, and far-right paired with center-left or far-left—exhibit coordination rates of 0.00 (±0.00) across all six runs. Even moderately distant pairings (e.g., center-left–center-right or neutral–far-right) rarely coordinate, with only isolated exceptions such as the Llama center-right–far-right dyad achieving a partial coordination rate of 0.50 (±0.00). When coordination does fail, the average first coordination round is undefined or effectively at the maximum horizon, reflecting the fact that agents never reach a coordinated outcome within the 25-round window. Crucially, this failure persists despite the incentive structure strongly favoring coordination over non-coordination: agents consistently forgo higher joint payoffs rather than concede to ideologically opposed partners.

These results demonstrate that monetary incentives alone are insufficient to induce coordination across ideological boundaries in persona-conditioned LLM agents. While aligned agents immediately exploit the payoff structure to coordinate, ideologically mismatched agents remain locked in non-coordinated outcomes, revealing a strong identity-based barrier that dominates instrumental reward maximization in the absence of social or identity-level communication

8.2.4 Discussing Shared Identity Promotes Coordination

When agents are allowed to discuss non-political aspects of their persona and share substantial overlap in social identity—such as age, gender, location, occupation, race, and hob-

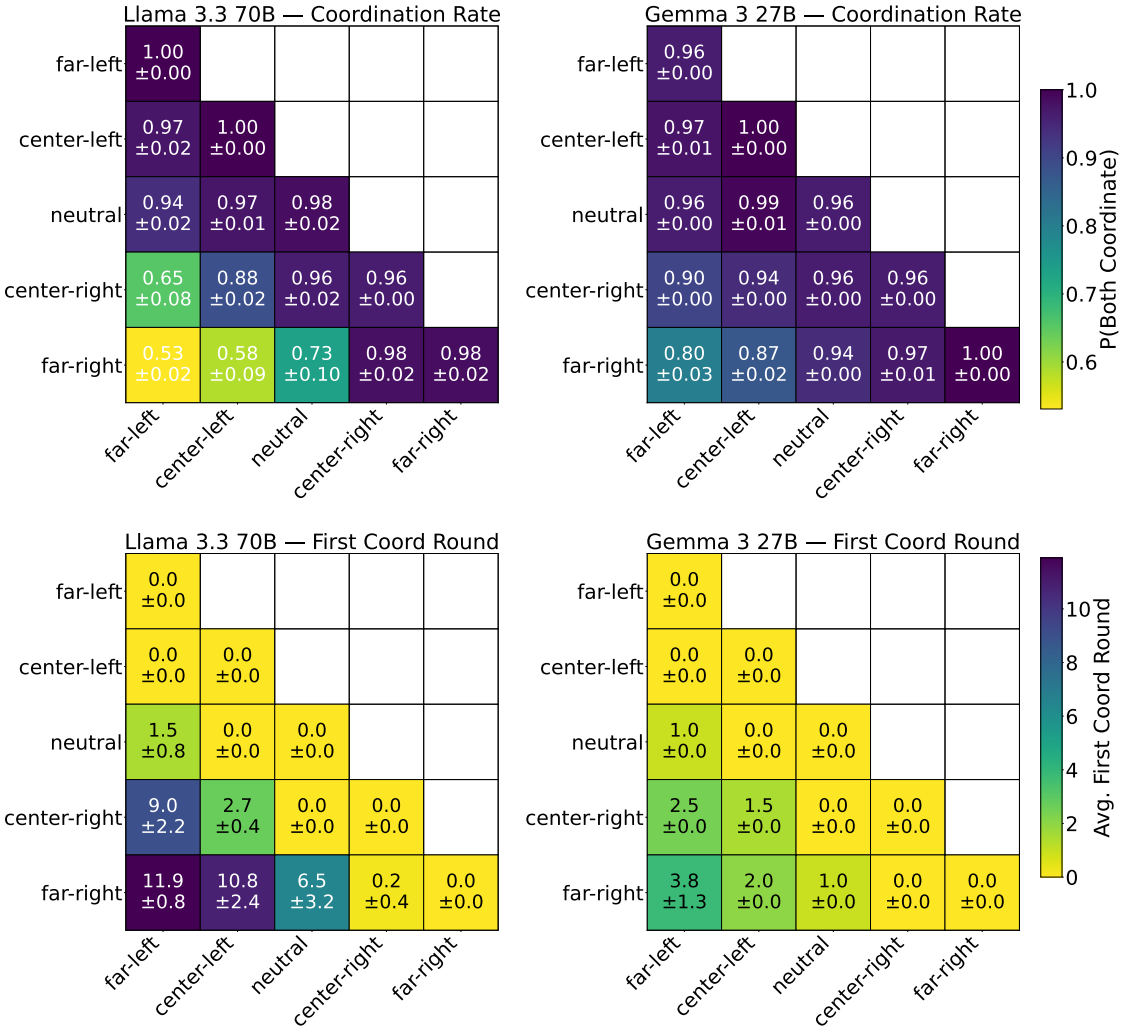

Figure 8.6: Coordination outcomes when persona-conditioned agents are allowed to communicate and have overlapping non-political identity traits. Heatmaps report mean coordination rates (top) and average first coordination round (bottom) across six dyadic runs for each ideological pairing.

bies—coordination rates increase dramatically across the ideological spectrum, consistent with a form of social identity priming (Figure 8.6). Unlike the baseline condition, where cross-ideological coordination was almost entirely absent, introducing shared identity cues substantially reduces ideological barriers to coordination for both models.

For **Llama-3.3-70B**, same-ideology dyads continue to coordinate at near-perfect rates (e.g., far-left–far-left: 1.00±0.00; far-right–far-right: 0.98±0.02), but critically, coordination emerges even among ideologically distant pairs. Far-left–center-right dyads coordinate in

0.65 ± 0.08 of runs, far-left–far-right dyads in 0.53 ± 0.02, and center-left–center-right dyads in 0.88 ± 0.02. Correspondingly, first coordination occurs at finite and interpretable times rather than never: center-right–far-left pairs coordinate after approximately 9.0 ± 2.2 rounds, while far-right–far-left pairs converge around 11.9 ± 0.8 rounds, indicating delayed but consistent coordination rather than persistent deadlock.

Gemma-3-27B exhibits an even stronger identity effect. Coordination rates exceed 0.90 for most dyads, including those spanning ideological extremes. Far-left–far-right pairs coordinate at 0.80 ± 0.03, center-left–far-right at 0.87 ± 0.02, and neutral–far-right at 0.94 ± 0.00. Time-to-coordination is correspondingly short: most cross-ideological dyads converge within the first 1–3 rounds, and many coordinate immediately (average first coordination round 0.0 ± 0.0). Even the slowest-converging pairs, such as far-right–far-left, coordinate by 3.8 ± 1.3 rounds.

When agents are allowed to communicate but share *no overlapping non-political identity traits* Figure 8.7, coordination improves relative to the no-communication baseline, but remains substantially slower and less reliable than in the shared-identity condition (Figure 8.6). In this setting, agents exchange information about their personas, but discover that they have no common ground outside ideology. As a result, any gains in coordination must arise from dialogue itself rather than from social identity overlap.

For **Llama-3.3-70B**, same-ideology dyads continue to coordinate at high rates (e.g., far-left–far-left: 0.98 ± 0.03; far-right–far-right: 0.94 ± 0.03), but coordination across ideological distance is attenuated and delayed. Moderately distant dyads such as center-left–neutral still coordinate frequently (0.95 ± 0.09), whereas ideologically opposed pairings show much lower rates: far-left–center-right coordinate at 0.48 ± 0.11, and far-left–far-right at 0.34 ± 0.11. Time-to-coordination increases sharply as ideological distance grows. Neutral–far-left dyads require 10.0 ± 2.7 rounds on average before first coordination, while far-left–center-right and far-left–far-right pairs converge only after approximately 16.8 ± 2.0 and 16.7 ± 2.5 rounds, respectively—approaching the full 25-round horizon.

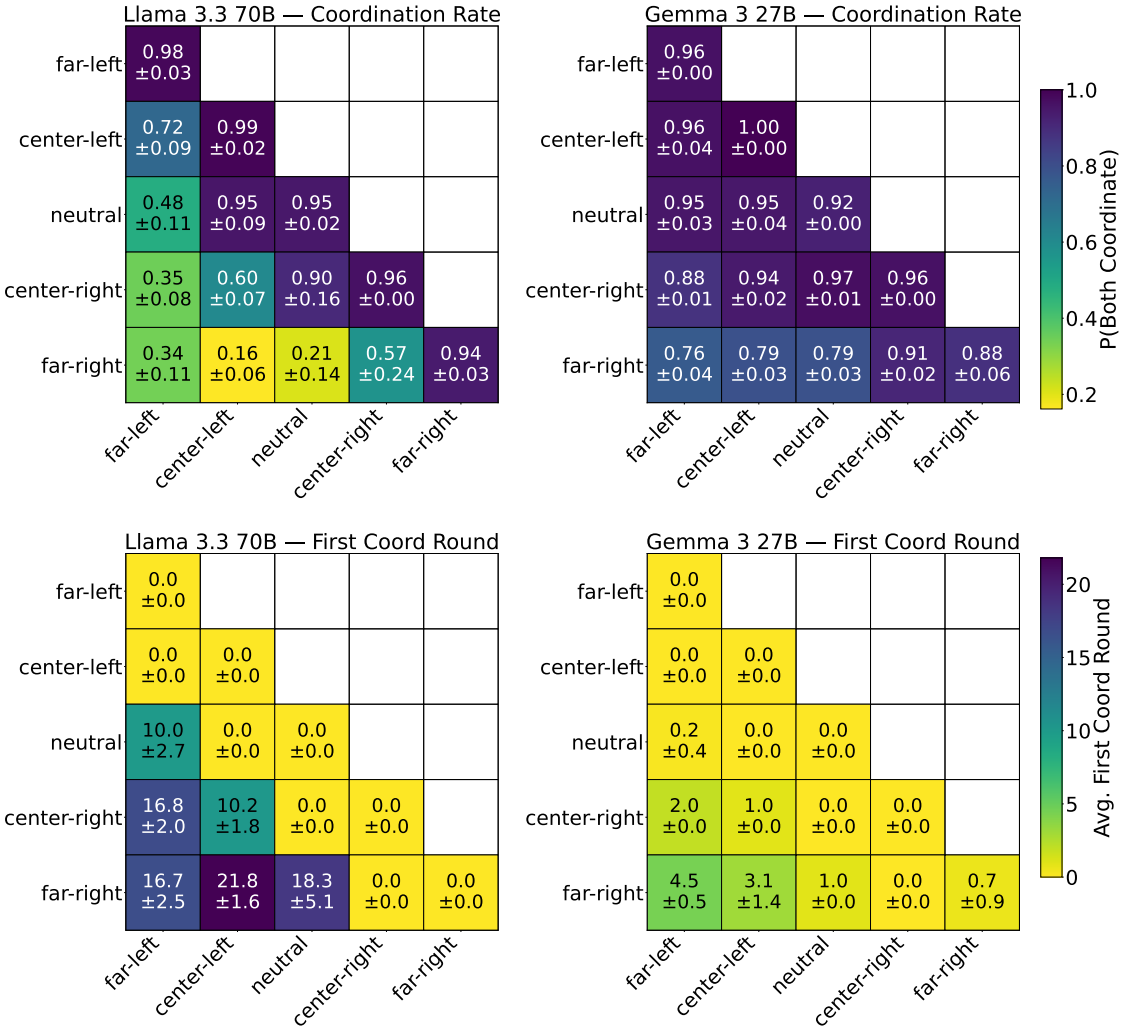

Figure 8.7: Coordination outcomes when persona-conditioned agents are allowed to communicate but have no overlapping non-political identity traits. Heatmaps report mean coordination rates (top) and average first coordination round (bottom) across six dyadic runs for each ideological pairing.

Gemma-3-27B exhibits the same qualitative pattern with generally higher coordination rates but clear temporal delays. While ideologically aligned dyads coordinate near perfectly (e.g., far-left–far-left: 0.96 ± 0.00; far-right–far-right: 0.88 ± 0.06), ideologically distant pairs converge more slowly. Far-left–far-right dyads coordinate at 0.76 ± 0.04, with first coordination occurring around 4.5 ± 0.5 rounds, while center-left–far-left pairs converge after 3.1 ± 1.4 rounds. Although coordination is typically achieved, it requires sustained interaction rather than immediate alignment.

Allowing persona-conditioned agents to communicate has a large but highly structured effect on coordination across ideological divides. When agents share substantial non-political identity overlap, coordination emerges rapidly and reliably even between ideologically distant pairs, with high coordination rates and short time-to-convergence across both Llama-3.3-70B and Gemma-3-27B. In contrast, when agents are allowed to communicate but share no overlapping identity traits, coordination still improves relative to the no-communication baseline, but occurs more slowly and less consistently, particularly for ideologically opposed dyads.